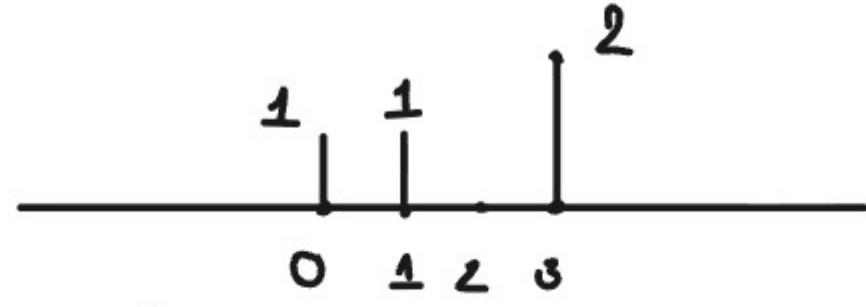


Lecture 3: DFT and Inverse DFT

$$X(\omega) = \frac{2\pi}{N} \sum_k X_o(k \frac{2\pi}{N}) \cdot \delta(\omega - k \frac{2\pi}{N})$$

Bài tập:



$$X_o(\omega) = \sum_{n=0}^3 x_o[n] \cdot e^{-j\omega n} = 1 + e^{-j\omega} + 2e^{-j2\omega}$$

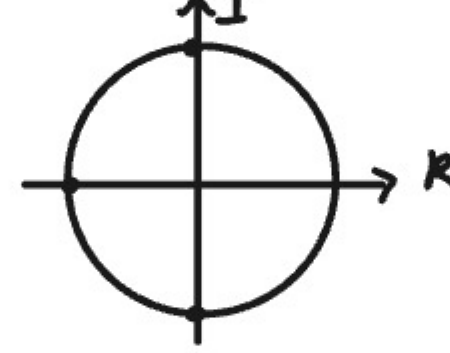
$$\text{Với } N=4, X_o(\frac{2\pi k}{4}) = 1 + e^{-j \frac{2\pi k}{4}} + 2e^{-j \frac{8\pi k}{4}} = 1 + e^{-j \frac{\pi k}{2}} + 2e^{-j 2\pi k}, k=0,1,2,3$$

$$+ k=0, X_o(0) = 1 + 1 + 2 = 4$$

$$+ k=1, X_o(1) = 1 + e^{-j \frac{\pi}{2}} + 2e^{-j \pi} = 1 - j + 2(-1) = 1 - j$$

$$+ k=2, X_o(2) = 1 + e^{-j \pi} + 2e^{-j 2\pi} = 1 + (-1) + 2(1) = 2$$

$$+ k=3, X_o(3) = 1 + e^{-j \frac{3\pi}{2}} + 2e^{-j \frac{9\pi}{2}} = 1 + j + 2(-1) = 1 - j$$



$$X(\omega) = \frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} X_o(\frac{2\pi k}{N}) \cdot \delta(\omega - \frac{2\pi k}{N}) = \frac{2\pi}{4} \sum_{k=0}^3 X_o(\frac{2\pi k}{4}) \delta(\omega - \frac{2\pi k}{4})$$

$$X(\omega) = \frac{\pi}{2} \left(4\delta(\omega) + (1-j)\delta(\omega - \frac{\pi}{2}) + 2\delta(\omega - \pi) + (1-j)\delta(\omega - \frac{3\pi}{2}) \right)$$

2) Windowing: $w_R = \begin{cases} 1, & n=0,1,\dots,N-1 \\ 0, & \text{còn lại} \end{cases}$

$$x_o[n] = x[n] \cdot w_R[n]$$

$$X_o(\omega) = \sum_{n=-\infty}^{+\infty} x_o[n] \cdot e^{-j\omega n} = \sum_{n=0}^{N-1} x[n] \cdot e^{-j\omega n}$$

+ Thừa số quay, số phức nằm trên đường tròn đơn vị.

$$W_N = e^{-j \frac{2\pi}{N}}$$

$$+ \text{DFT: } X[k] = \sum_{n=0}^{N-1} x[n] \cdot W_N^{kn}$$

$$+ \text{IDFT: } x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \cdot W_N^{-kn}$$

Bài tập: Tìm DFT của $x[n] = 0, 1, 2, \dots, N-1$

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn} = \sum_{n=0}^{N-1} n W_N^{kn} = \frac{1 - W_N^{kN}}{1 - W_N^k}$$

$$k=0 \Rightarrow X[k] = X(0) = N$$

$$k \neq 0 \Rightarrow X[k] = 0$$

$$X[k] = N \delta[k]$$

Bài tập: Tìm DFT của

$$a) x[n] = \delta[n] = \begin{cases} 1 & \text{với } n=0 \\ 0 & \text{với } n \neq 0 \end{cases}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot W_N^{kn} = \sum_{n=0}^{N-1} \delta[n] \cdot e^{-j \frac{2\pi kn}{N}} = e^{-j \frac{2\pi k \cdot 0}{N}} = 1$$

$$b) x[n] = a^n \cdot \{u[n] - u[n-N]\}$$

$$x[n] = \begin{cases} a^n & \text{với } 0 \leq n \leq N-1 \\ 0 & \text{còn lại} \end{cases}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot W_N^{kn} = \sum_{n=0}^{N-1} a^n \cdot e^{-j \frac{2\pi kn}{N}} = \sum_{n=0}^{N-1} (a \cdot e^{-j \frac{2\pi k}{N}})^n$$

$$X[k] = \frac{1 - (a \cdot e^{-j \frac{2\pi k}{N}})^N}{1 - a \cdot e^{-j \frac{2\pi k}{N}}} = \frac{1 - a^N \cdot e^{-j 2\pi k}}{1 - a \cdot e^{-j \frac{2\pi k}{N}}} = \frac{1 - a^N}{1 - a \cdot e^{-j \frac{2\pi k}{N}}}$$

$$c) x[n] = \begin{cases} 1 & \text{nếu } n \text{ chẵn} \\ 0 & \text{nếu } n \text{ lẻ} \end{cases}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot W_N^{kn} = \sum_{n=0}^{N-1} \delta[n \text{ chẵn}] \cdot e^{-j \frac{2\pi kn}{N}}$$

Đặt $n=2m$, ta được

$$X[k] = \sum_{m=0}^{\frac{N}{2}-1} x[2m] \cdot e^{-j \frac{2\pi k \cdot 2m}{N}} = \sum_{m=0}^{\frac{N}{2}-1} 1 \cdot e^{-j \frac{4\pi km}{N}} = \sum_{m=0}^{\frac{N}{2}-1} (e^{-j \frac{4\pi k}{N}})^m$$

$$\text{TH 1, khi } k \text{ là bội số của } \frac{N}{2} \text{ thì } e^{-j \frac{4\pi k}{N}} = 1$$

$$\Rightarrow X[k] = \frac{N}{2}$$

$$\text{TH 2, } X[k] = \frac{1 - (e^{-j \frac{4\pi k}{N}})^{\frac{N}{2}}}{1 - e^{-j \frac{4\pi k}{N}}} = \frac{1 - e^{-j 2\pi k}}{1 - e^{-j \frac{4\pi k}{N}}} = 0$$

$$\text{Vậy } X[k] = \begin{cases} \frac{N}{2} & \text{khi } k=0, \frac{N}{2} \\ 0 & \text{còn lại} \end{cases}$$

Bài 2:

Tính chất dịch vòng của DFT

$$x[n-m] \xrightarrow{\text{DFT}} W_N^{km} \cdot X[k]$$

$$x[n] = [1; 0,75; 0,5; 0,25]$$

$$y[n] = [0,75; 0,5; 0,25; 1]$$

Dãy $y[n]$ chính là $x[n]$ dịch sang trái 1 mẫu

$$y[n] = x[n+1]$$

$$\text{Mà } x[n-m] \xleftrightarrow{\text{DFT}} W_N^{mk} \cdot X[k]$$

$$\Rightarrow Y[k] = W_N^{-1k} \cdot X[k]$$

$$= e^{-j \frac{2\pi k(-1)}{4}} \cdot X[k]$$

$$= e^{j \frac{\pi k}{2}} \cdot X[k]$$